

# Ask Google Before Statistics: Nowcasting Unemployment in Turkey Using Google Trends\*

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## Abstract

Official unemployment statistics for Turkey are published with a lag of roughly two months, leaving policymakers with a ragged-edge information gap precisely when labor-market conditions move fastest. We ask whether online search behavior, available in real time, can be used to nowcast the current unemployment rate before its official release. We assemble a panel of 21 Turkish labor-market search queries (2010–2025) and calibrate them with the Google Trends Anchor Bank (GTAB), which recovers low-volume signals that raw Google Trends rounds to zero. We summarize the calibrated panel with a small set of varimax-rotated principal components and embed them in a state-space model whose latent labor-market component is tracked by the Kalman filter, with lagged search factors entering the unemployment equation as a bridge. We evaluate the model in a recursive pseudo-out-of-sample exercise over 2018–2025 that masks the two-month publication lag, and we compare its accuracy against autoregressive benchmarks using the Diebold–Mariano and Clark–West tests. Augmenting the benchmark with Google Trends lowers the root mean squared error by up to roughly 9% over the full window, and by nearly 20% before the pandemic; the improvement is significant under the Clark–West test for nested models. The gain is not constant, however: it is concentrated in the 2018–2021 part of the sample and fades to essentially zero after 2022, so the predictive content of search behavior decays over time rather than switching on only in crises. A real-time rule that re-selects the model each year still beats the benchmark, but a model-confidence-set analysis shows that, at this sample size, no specification can be statistically separated from the rest.

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\*Replication code and data are available at <https://github.com/borderedhessian33/EC432-Google-Unemployment>.

**Keywords:** nowcasting; Google Trends; unemployment; Kalman filter; state-space models; Turkey.

**JEL codes:** C53, E24, E27, J64.

# 1 Introduction

Macroeconomic decisions are made in real time, but the data that describe the economy arrive late. In Turkey, the Turkish Statistical Institute (TÜİK) releases the unemployment rate for month  $t$  only at the end of month  $t + 2$ , so at any point in time the most recent official figure already describes conditions one to two months in the past. This ragged edge is most costly exactly when it matters most: during recessions, currency crises and pandemics, labor-market conditions can deteriorate sharply long before they leave any trace in the published series. Nowcasting—estimating the current state of the economy from information available today—has therefore become a central tool of applied macroeconomics (Giannone, Reichlin, and Small, 2008), and high-frequency digital footprints such as web-search activity are among its most promising inputs (Choi and Varian, 2012).

Search data are attractive for this purpose because job seekers reveal their intentions online before those intentions show up in administrative statistics, and because the signal is available essentially without delay. Turkey is an especially favorable setting: Google handles roughly 97% of all searches in the country, so query volumes offer broad and high-quality coverage of labor-market interest. Yet raw Google Trends is not directly usable as an econometric input. It returns a relative index that is rescaled within each batch request, so the same keyword takes different values across batches, and it rounds low-volume queries to zero, discarding exactly the niche search terms that may carry the most specific labor-market information. We address this with the Google Trends Anchor Bank (GTAB), which anchors every query to a common reference term and applies a log-linear calibration across overlapping windows, producing a consistent panel of search intensities that is comparable across keywords (West, 2020).

This paper studies whether such a GTAB-calibrated search panel improves real-time inference about Turkish unemployment. We construct a composite search index from 21 labor-market keywords using a varimax-rotated principal-component analysis, embed it in a state-space model whose latent labor-market factor is estimated by the Kalman filter, with lagged search factors entering the unemployment equation through a bridge. We then ask a sharp, testable question: does augmenting an autoregressive benchmark with Google Trends reduce the out-of-sample forecast error, especially during periods of heightened uncertainty?

Our contribution is empirical. To our knowledge, this is the first nowcasting model of Turkish unemployment built on GTAB-calibrated Google Trends, estimated over a long

2010–2025 sample that spans the 2018 currency crisis and the COVID-19 pandemic. We embed a composite search index in a state-space model in which unemployment loads both on a shared latent factor and on lagged search factors through a bridge, and we evaluate the model in a genuine real-time, pseudo-out-of-sample exercise: the preprocessing pipeline is frozen before the evaluation period and the two-month publication lag is masked, so that the reported accuracy reflects only information that would have been available at the time. This design lets us measure the value of search data cleanly and to ask not merely whether Google Trends help, but when.

The results paint a nuanced picture that we believe is the paper’s main substantive message. Adding Google Trends lowers the root mean squared error relative to an AR(1) benchmark by up to roughly 9% over the full window, and the improvement is significant under the Clark–West test. The gain is far from uniform over time, however: it is large through the 2018 currency crisis and the COVID-19 pandemic and then fades to essentially nothing after 2022, so the predictive content of search behavior appears to decay over the sample rather than to switch on only in unstable times. A real-time rule that re-selects the model each year still beats the benchmark, yet a model confidence set shows that, with only eight years of monthly data, no specification—search-augmented or not—can be told apart from the rest. The honest reading is that search data deliver a real but modest and time-limited edge, largest exactly when official statistics are most stale.

The remainder of the paper is organized as follows. Section 2 reviews the literature on nowcasting with search data and locates our contribution within it. Section 4 describes the data, the GTAB calibration, the preprocessing pipeline and the state-space nowcasting model. Section 5 reports the recursive out-of-sample results and the predictive-accuracy tests, and Section 6 discusses their interpretation. Section 7 concludes with limitations and directions for future researchs.

## 2 Literature Review

**Predicting the present with search data.** The idea that web-search activity can track the economy in real time predates Google Trends itself. An influential early demonstration came from epidemiology: Ginsberg et al. (2009) tracked influenza outbreaks in near real time from the volume of related search queries, and the same idea was soon carried into economics. Ettredge, Gerdes, and Karuga (2005) showed that web-based search queries contained information about official labor-market statistics, and Askittas and Zimmermann (2009) demonstrated that Google query series could forecast German unemployment. The literature was consolidated by Choi and Varian (2012), whose “predicting the present” agenda found that Google search volume improves near-term forecasts of a wide range of indicators, from unemployment claims to auto sales and

consumer confidence. Varian (2014) and Scott and Varian (2014) formalized how such high-frequency signals can be folded into structural time-series forecasts, and Vosen and Schmidt (2011) show that Google-based indicators can outperform survey-based ones for private consumption. Beyond search data, Blanchflower and Bryson (2023) use European Commission survey data for 29 countries over 1985–2022, an unbalanced panel of roughly 10,000 country-month observations, to predict changes in the unemployment rate twelve months ahead. They find that individuals’ fears of unemployment predict subsequent changes in unemployment twelve months later, and that industrial firms’ employment expectations are predictive three months ahead, with out-of-sample forecasts that track actual unemployment through two major recessions.

**Forecasting unemployment with Google Trends.** A more focused strand evaluates search data specifically for unemployment. D’Amuri and Marcucci (2017) construct a Google job-search index for the United States and show that it outperforms autoregressive benchmarks in forecasting the unemployment rate, with predictive gains that are robust across pre-, during- and post-recession periods. Borup and Schütte (2022) exploit a large, 172-variable Google Trends panel for US employment and, using Elastic Net and random forests, obtain out-of-sample  $R^2$  values of 29–62% at horizons of one to twelve months. Mulero and Garcia-Hiernaux (2023) use Google Trends job-search queries to forecast unemployment in Spain and study the digital divide by age and gender, finding that Google Trends queries are considerably better predictors of female than of male unemployment and that a larger set of useful queries is identified for women. Closely related single-country studies reach broadly positive but heterogeneous conclusions: Smith (2016) pairs Google Trends with mixed-frequency (MIDAS) regressions to nowcast UK unemployment, Pavlicek and Kristoufek (2015) document gains for the Visegrád economies, and Naccarato et al. (2018) combine official and search data to forecast the Italian youth unemployment rate. Baker and Fradkin (2017) build a Google-based job-search index and use it to study how job-search effort responds to unemployment-insurance policy, finding that search intensity rises as benefits approach exhaustion and is lower when benefits are more generous. Henson (2025) forecasts US payroll employment in real time with Google Trends embedded in dynamic factor models benchmarked against an ARMA(1,1) specification; evaluating forecasts on data as it was originally released, he finds virtually no gain from Google Trends for national employment and only small, uneven gains across individual states, in contrast to studies that rely on the fully revised history.

**Evidence for Turkey.** The work closest to ours is Chadwick and Şengül (2015), who nowcast the Turkish unemployment rate by combining 35 Google queries through Bayesian model averaging and principal components under a three-month lag structure, reporting substantial in-sample accuracy gains. Their study is our direct methodological

template, but we depart from it in three consequential ways. We evaluate performance out of sample rather than in sample; we calibrate the raw search panel with GTAB before any modelling, recovering low-volume signals that batched Google Trends suppresses; and rather than combining the search factors through model averaging of single-equation regressions, we introduce an explicit latent labor-market factor and embed it in a state-space model estimated by the Kalman filter.

**Calibrating Google Trends.** A practical obstacle common to all of these studies is that raw Google Trends is a noisy, batch-dependent measurement of search intensity. West (2020) introduces the Google Trends Anchor Bank (GTAB), which anchors all queries to a common reference term and corrects rounding and rescaling artefacts through a log-linear calibration across overlapping time windows, yielding a panel of search intensities that is comparable across keywords.

## 3 Data

### 3.1 Unemployment and Google Series

We will proceed by introducing the data, we use the Turkish monthly unemployment series between 2010-2025 pulled from the EVDS system of the Turkish Central Bank. The data has been already deseasonalized thus we do not implement any such procedure. As seen in figure 1 Turkish unemployment has been hovering around 7.9 percent to 14.1 percent within the 15 year span. Especially during the 2018 currency crisis and COVID 2020 pandemic unemployment numbers have been maintained between 10 to 14 percent a high for the period. However in the last 2 years it has stabilized to more or less around 8 percent, indicating the presence of a calmer period.

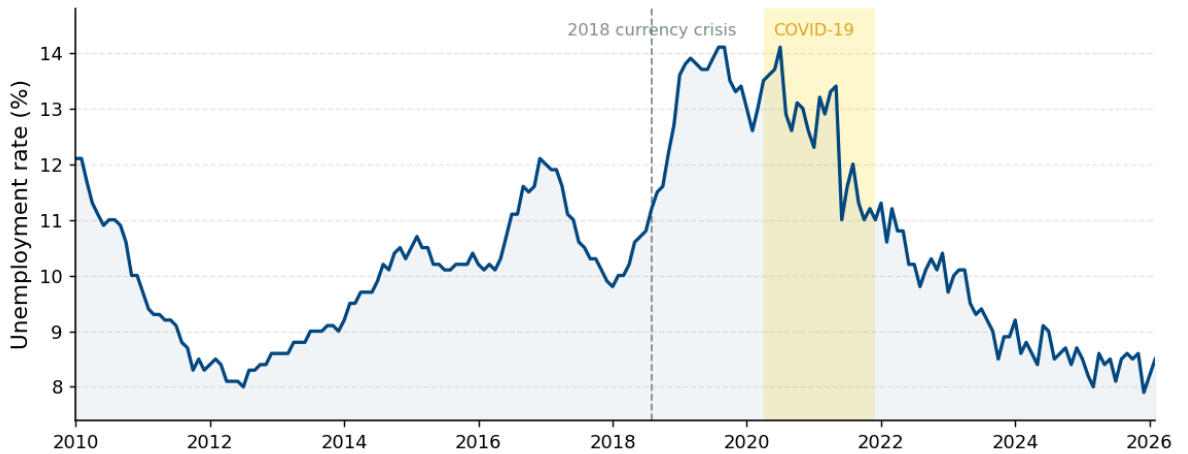


Figure 1: Turkish unemployment series, 2010–2025

On the side of google trends we have 21 job search related keywords that are mainly inspired by the literature. All search words that are used are calibrated versions pulled via API using the G-TAB package. To show the importance of calibration figure 2 shows the series for "işkur", "sgk işten çıkış", "işsizlik başvuru" and "ihbar tazminatı". As it can be seen due to the presence of a term that is searched intensely-that being işkur- the other 3 terms collapse to 0. When we use G-TAB to calibrate our dataset, we see that these three terms attain their real values and their respective trends.

In more detail the keywords we selected cover the potential cycle a labor market could go through. There are potential terms that people may search during job loss such as "işten çıkarılma tazminatı" or "kıdem tazminatı hesaplama", active job search ones("iş ilanı", "iş başvuru", "memur alımı") and potential terms people search when benefit seeking like "işsizlik maaşı", "işsizlik ödeneği" or "işkur". Some terms are more real time indicators of unemployment with "işsizlik" and "sgk işten çıkış" while some terms may be lagged indicators of unemployment distress. In other words, employees may feel that they could be dismissed hence they search for aforementioned job loss terms to hedge themselves against such scenario.

Table 1: Google Trends Keywords Used in the Analysis

| #  | Keyword (Turkish)             | Translation                       | Category      |
|----|-------------------------------|-----------------------------------|---------------|
| 1  | işsizlik                      | unemployment                      | General       |
| 2  | işsizlik maaşı                | unemployment benefit              | Benefits      |
| 3  | işsizlik ödeneği              | unemployment allowance            | Benefits      |
| 4  | işsizlik başvurusu            | unemployment application          | Benefits      |
| 5  | işsizlik ödeneği nasıl alınır | how to get unemployment allowance | Benefits      |
| 6  | işkur                         | Turkish Employment Agency (İŞKUR) | Institutions  |
| 7  | işkur başvuru                 | İŞKUR application                 | Institutions  |
| 8  | sgk işten çıkış               | SGK job exit (social security)    | Institutions  |
| 9  | işten çıkarılma tazminatı     | dismissal compensation            | Severance     |
| 10 | ihbar tazminatı               | notice pay                        | Severance     |
| 11 | kıdem tazminatı hesaplama     | severance pay calculation         | Severance     |
| 12 | iş ilanı                      | job posting                       | Job Search    |
| 13 | iş arama                      | job search                        | Job Search    |
| 14 | iş başvurusu                  | job application                   | Job Search    |
| 15 | özgeçmiş                      | résumé / CV                       | Job Search    |
| 16 | kariyer.net                   | kariyer.net (job portal)          | Job Search    |
| 17 | mülakat soruları              | interview questions               | Job Search    |
| 18 | ek iş                         | extra / side job                  | Job Search    |
| 19 | memur alımı                   | civil servant recruitment         | Public Sector |
| 20 | fabrika işçi alımı            | factory worker recruitment        | Public Sector |
| 21 | güvenlik görevlisi alımı      | security guard recruitment        | Public Sector |

Selected 21 job search terms with their respective category.

Figure 3 also shows the time series of selected job search terms. Some terms retain their high search frequency due to their timeless nature such as işkur, işsizlik maaşı or iş ilanı, where as kariyer.net suffers a huge drop-off in 2013 due to losing it's popularity to other alternative job search platforms like LinkedIn. We see that google searches are disturbed in times of precarity such as the COVID outbreak although the direction and magnitude of the response differ across keywords. Shorter spikes of "işkur" and "iş alımı" may reflect administrative announcements or application deadlines rather than movements in unemployment alone. Especially in "memur alımı" these short lived spikes can be a result of public-sector recruitment cycles. As observed there is considerable heterogeneity within the movements of these job-search terms. This signals to us that there may not be a single common signal, thus it will be methodologically beneficial to approach within a factor extraction setup such as PCA or PLS. This also supports treating Google variables as noisy measurements of an underlying latent labor-market condition.

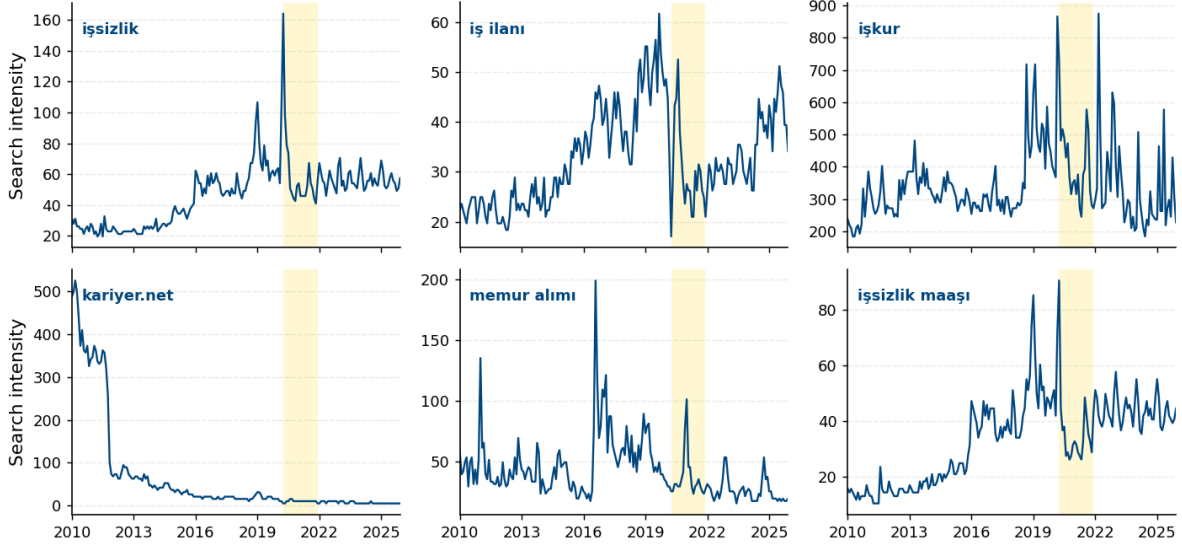


Figure 2: Selected Google Trends Search Terms 2010–2025

### 3.2 Calibrating with GTAB

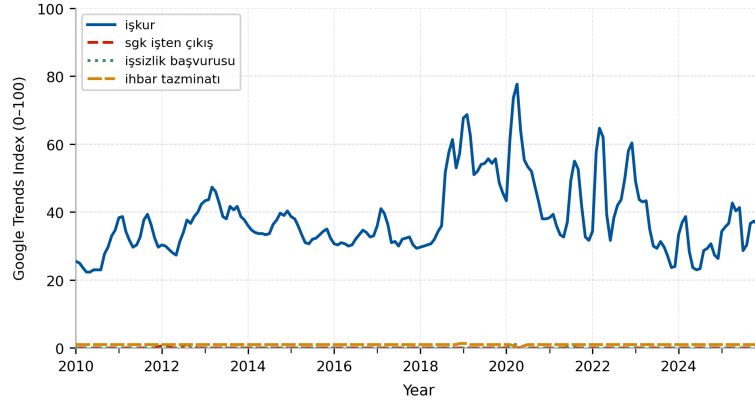
Raw Google Trends is a convenient but difficult data source to interpret, and three of its properties make it unusable as a direct econometric input. First, it reports a relative index on a 0–100 scale rather than absolute search volume: the figures describe the popularity of a query relative to its own history, not the number of searches it received. Second, Google Trends accepts only a handful of queries per request and rescales each request so that the most popular query in that batch equals 100; the same keyword therefore takes different values depending on which other keywords it is requested alongside, so series pulled from different batches are not comparable. Third, and most damaging for our purposes, low-volume queries are rounded down—often all the way to zero—when they are requested together with a dominant term, so that genuinely informative niche queries are recorded as flat lines of zeros.

These properties are problematic for a study like ours that combines many keywords. The principal-component analysis below presumes that its inputs are measured on a common scale and carry real variation; with raw Trends, a few high-volume queries would mechanically dominate the components, batch-dependent rescaling would inject artificial level shifts across the 2010–2025 sample, and the zeroed-out low-volume terms would contribute no information at all. The Google Trends Anchor Bank (GTAB; West, 2020) is designed precisely to remove these distortions. GTAB first builds an anchor bank: a ladder of reference queries that span many orders of magnitude of search volume and that are placed on a single common scale by querying them in overlapping batches. Any target keyword is then matched to an anchor of comparable magnitude and rescaled onto that global scale, with a log-linear calibration across overlapping windows that undoes the rounding. The output is a panel in which every keyword is expressed in the same units,



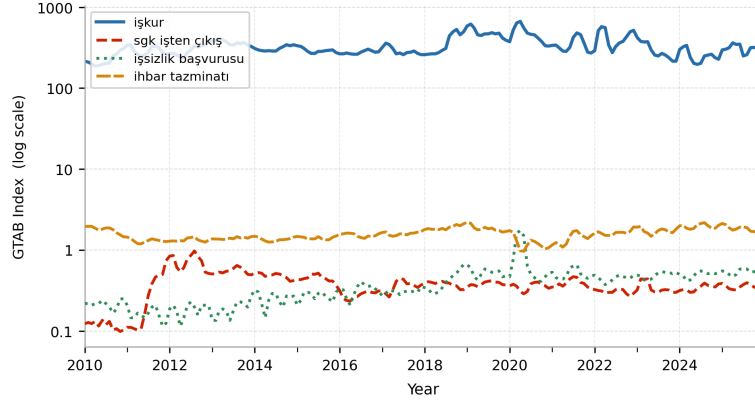
is comparable both across keywords and over time, and retains the non-zero variation of low-volume queries.

The practical payoff is visible in our own data. Figure 3 contrasts the raw and GTAB-calibrated series for three low-volume Turkish labor-market queries—sgk işten çıkış, işsizlik başvurusu and ihbar tazminatı—when they are requested alongside the dominant anchor işkur. In the raw panel the three niche queries are largely flattened to zero and swamped by the anchor; after calibration they recover clearly distinguishable, economically meaningful movements. We apply GTAB to all 21 keywords over 2010–2025, producing the calibrated panel used throughout the rest of the paper.



Source: Google Trends (raw, 0–100 scale). 3-month rolling mean applied.

(a) Raw Google Trends



Source: Google Trends via GTAB calibration. 3-month rolling mean applied.

(b) GTAB-calibrated

Figure 3: Effect of GTAB calibration on low-volume search queries. Both panels plot the anchor query işkur together with three low-volume queries (sgk işten çıkış, işsizlik başvurusu, ihbar tazminatı) over 2010–2025, with a three-month rolling mean and the year on the horizontal axis. Panel (a) shows the raw Google Trends index (0–100 scale), on which the three low-volume queries are flattened to near zero and dwarfed by the anchor; panel (b) shows the GTAB-calibrated index on a logarithmic scale, on which the same three queries recover clearly distinguishable, non-zero variation. Source: Google Trends, raw and via GTAB calibration.

## 4 Methodology

### 4.1 PCA and Data Preparation

To extract each signal from the google trend variables we employ principal components analysis, the standard tool for distilling many noisy predictors into a few common factors in macroeconomic forecasting (Stock and Watson, 2002; Forni et al., 2000). First for each keyword, let  $x_{it}$  denote the GTAB-calibrated search index for keyword  $i \in \{1, \dots, 21\}$  at month  $t$ . To remove seasonality (as these keywords contain their seasonal component) and induce some form of stationarity, we apply a twelve-month year-on-year difference:

$$\tilde{x}_{it} = x_{it} - x_{i,t-12}. \quad (1)$$

After this ch differenced series is then standardised using the mean  $\bar{\tilde{x}}_i$  and standard deviation  $s_i$  exclusively on the training window  $t \leq T_0$  (including December 2017):

$$z_{it} = \frac{\tilde{x}_{it} - \bar{\tilde{x}}_i}{s_i}, \quad \bar{\tilde{x}}_i, s_i \text{ estimated for } t \leq T_0. \quad (2)$$

Freezing the normalisation at  $T_0$  prevents any look-ahead bias into the recursive out-of-sample setup we will employ when forecasting.<sup>1</sup>

Following this step let the standardised keywords form into the  $\mathbf{z}_t$ . The sample covariance matrix on the training window is

$$\hat{\Sigma} = \frac{1}{T_0} \sum_{t=1}^{T_0} \mathbf{z}_t \mathbf{z}_t', \quad \hat{\Sigma} \in \mathbb{R}^{21 \times 21}. \quad (3)$$

The correlation matrix of the keywords can be found in figure 4. Via PCA we extract the  $K = 4$  leading eigenvector pairs so the first 4 PCs.  $(\hat{\lambda}_k, \mathbf{v}_k)$  satisfying  $\hat{\Sigma} \mathbf{v}_k = \hat{\lambda}_k \mathbf{v}_k$ , with  $\hat{\lambda}_1 \geq \dots \geq \hat{\lambda}_4 > 0$ . The raw score for component  $k$  is  $p_{kt} = \mathbf{v}_k' \mathbf{z}_t$ . The four components together account for nearly 60 percent of the shared variation across the 21 variables. This might be caused by the weakish correlation matrix as the variables individually contained rich signals, but as mentioned there is the presence of heterogeneity between them, which causes the individual PC's to explain the cumulative variance sparsely.

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<sup>1</sup>The twelve-month difference already eliminates a monthly seasonal pattern, so an additional seasonal adjustment step is not necessary; in some models we employed both STL decomposition and the YoY differences it proved that the results were very close to the only YoY case so we decided to proceed with the latter.

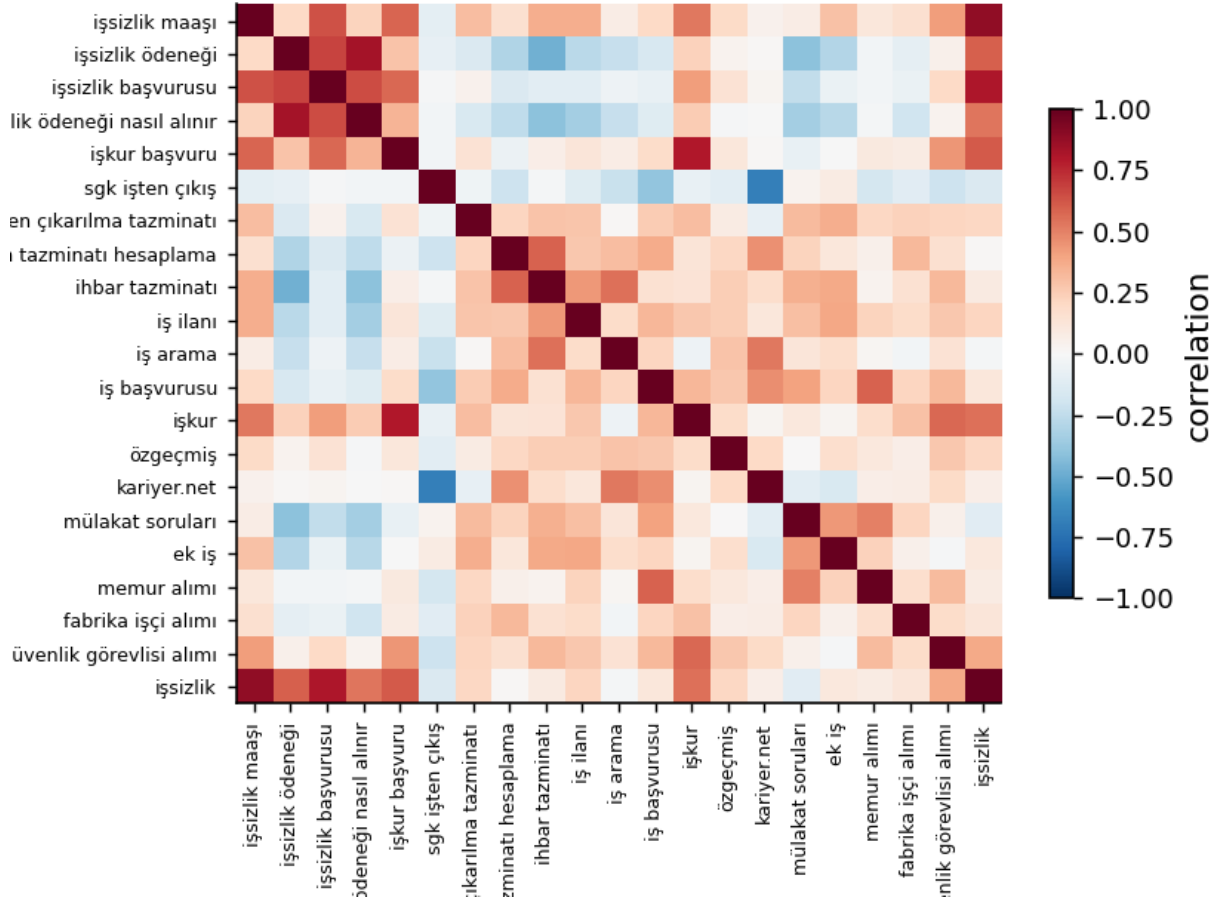


Figure 4: Correlation Matrix of Google Trend Variables

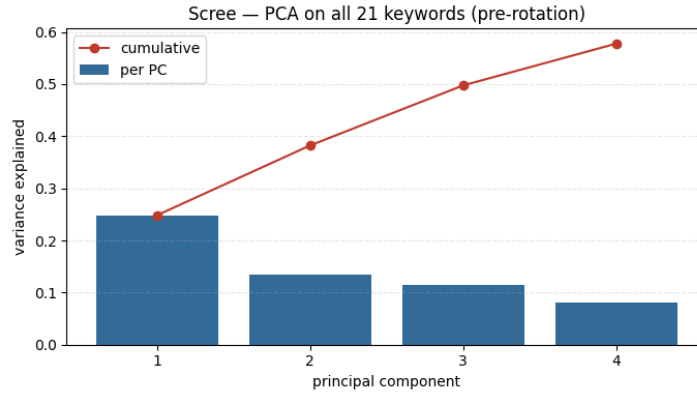


Figure 5: Scree Plot of Leading 4 PCs

Even though the raw PCA components maximise the explained variance, their loadings are diffused through each keyword so they may load moderately on every component. Because of this to obtain interpretable factors, we apply the varimax criterion. We therefore apply an orthogonal varimax rotation (Kaiser, 1958), which redistributes the loadings so that each component is more strongly associated with a smaller set of keywords. Because the rotation we do is orthogonal, it preserves the total variance explained by the

retained components while improving their interpretability.

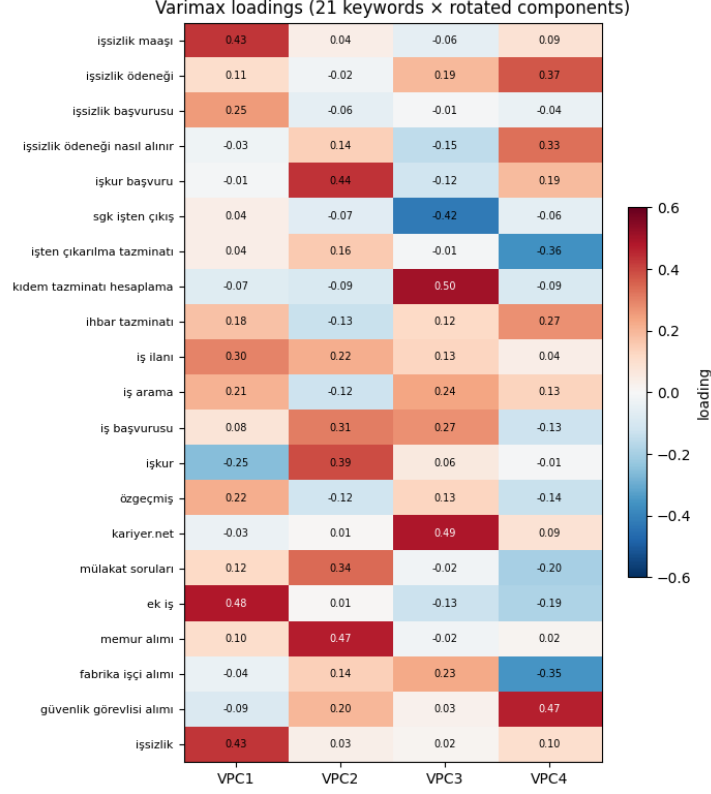


Figure 6: Loadings of the first 4 PC's after Varimax rotation

The signs of these newly rotated components are arbitrary. Because of this we multiply each component by  $-1$  whenever its correlation with unemployment is negative in our training sample. This ensures that higher values can consistently indicate weaker labor-market conditions:

$$\widehat{VPC}_{kt} = \text{sign}(\hat{\rho}_k) f_{kt}, \quad (4)$$

where  $\hat{\rho}_k$  is the training-sample correlation between component  $k$  and the unemployment rate. We then define the main labor-market factor as the component with the strongest correlation with unemployment:

$$\ell^* = \arg \max_{k=1, \dots, K} \hat{\rho}(\widehat{VPC}_k, u). \quad (5)$$

Out of the first 4 PC's, VPC1, VPC2, VPC3 and VPC4 respectively have a correlation of 0.48, 0.40, 0.61 and 0.07 with unemployment. Which indicates that VPC3 is the main labor market factor as define by us. In the apendx you can find the time series of each of the VPC's. To check whether VPC3 leads or lags unemployment we look at it's cross correlation with unemployment at different lags. In figure 7 we observe that VPC3 attains the highest cross correlation with unemployment at lag 2. Which is consistent with the

initial premise we started with, that being unemployment is reflected in a two month lag. So VPC3 our leading google trends factor is at best 2 months ahead of the unemployment statistic.

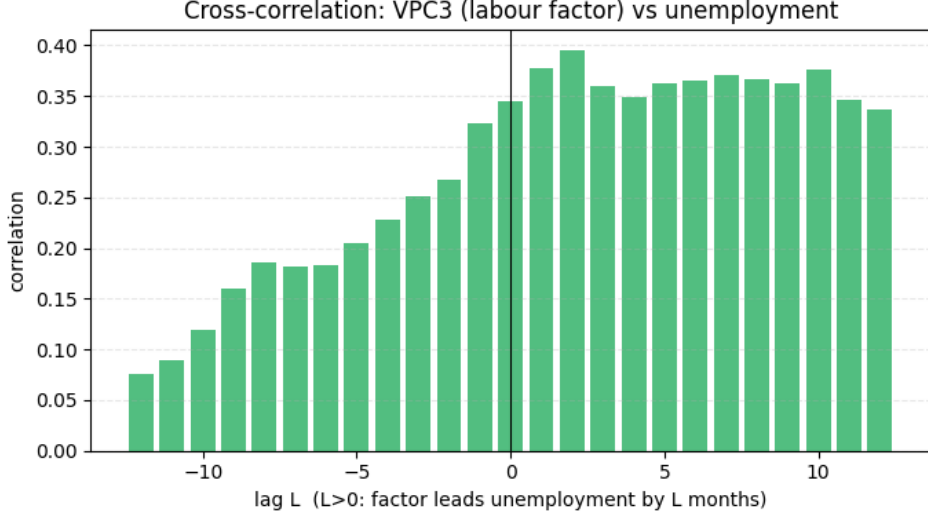


Figure 7: Cross Correlation of VPC3 with unemployment

## 4.2 The State-Space Nowcasting Model

All of our specifications share a common state-space structure and differ only in a handful of choices, which we leave free to explain here and focus the details more on the model selection. The four varimax components  $VPC_1, \dots, VPC_4$  constructed above are included in a subset  $\mathcal{S} \subseteq \{1, 2, 3, 4\}$  of them. A single latent monthly factor  $\mu_t$  of the underlying labor-market state determines both the observed unemployment rate  $u_t$  and the VPC's(search factors).

The latent factor is modeled as an AR process of order  $p$ ,

$$\mu_t = \sum_{i=1}^p \phi_i \mu_{t-i} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma_\eta^2), \quad (6)$$

The measurement equations combines  $u_t$  and the retained VPC's. The unemployment equation links  $u_t$  to the contemporaneous latent factor and to a bridge of lagged search factors, and to a COVID dummy(in some models not in all of them):

$$u_t = \mu_t + \sum_{j \in \mathcal{S}} \sum_{\ell=1}^L \beta_{j,\ell} VPC_{j,t-\ell} + \delta_t D_t^{\text{cov}} + \varepsilon_{u,t}, \quad \varepsilon_{u,t} \sim \mathcal{N}(0, \sigma_u^2), \quad (7)$$

while each retained search factor loads on current and lagged values of the latent state,

$$\text{VPC}_{j,t} = \sum_{d=0}^D \lambda_{j,d} \mu_{t-d} + \varepsilon_{j,t}, \quad \varepsilon_{j,t} \sim \mathcal{N}(0, \sigma_j^2), \quad j \in \mathcal{S}. \quad (8)$$

The disturbances  $\eta_t$ ,  $\varepsilon_{u,t}$  and  $\varepsilon_{j,t}$  are mutually and serially independent of each other.

We observe that the factors enter through two channels. Equation (8) is the shared-factor channel: because  $u_t$  and each  $\text{VPC}_{j,t}$  loads on the latent state  $\mu_t$ , the Kalman filter extracts labor-market information from the contemporaneously observed search factors before  $u_t$  is released, which is what enables nowcasting under the publication lag of 2. Equation (7) adds the bridge channel, in which the  $L$  lagged factors shift  $u_t$  directly, capturing the search activity that leads unemployment before conventional statistics detect them.

The system (6)–(8) is linear and Gaussian. The latent state is filtered with the Kalman filter (Kalman, 1960) and the parameters

$$\boldsymbol{\theta} = \left( \{\phi_i\}, \{\lambda_{j,d}\}, \{\beta_{j,\ell}\}, \delta, \sigma_\eta^2, \sigma_u^2, \{\sigma_j^2\}, \sigma_\zeta^2 \right)$$

are estimated by maximum likelihood (Durbin and Koopman, 2012) using the Kalman filter, subject to the usual stationarity and positivity constraints.

We define each specification to be indexed by the factor set  $\mathcal{S}$ , the autoregressive order  $p$ , the loading-lag  $D$ , the bridge-lag  $L$ , and whether the COVID term is absent or constant, . Setting  $\sigma_u^2 \rightarrow 0$  collapses the model to a pure latent-factor decomposition, and  $\mathcal{S} = \emptyset$  reduces it to an  $\text{AR}(p)$  benchmark on  $u_t$ ; both serve as nested comparisons later. We fix  $(p, D, L) = (1, 2, L)$  from the the two-month publication horizon and to retain statistical power that would have been lost with higher order lags of the the transition equation.

### 4.3 Real-Time Out-of-Sample Design

We evaluate every specification in a recursive out-of-sample forecasting setup that tries to mimic the available information a forecaster would actually have had in real time (Bańbura et al., 2013). The preprocessing pipeline follows as year-on-year differencing, PCA, and the varimax rotation. This is estimated up until December 2017 and then held fixed, so the rotated factors  $\text{VPC}_{j,t}$  always come from frozen loadings. It would have been more optimal and better as a design to rerun the PCA at each  $t$  however it would have been computationally laborious and the gains would have been relatively minimal. As time moves on the only parameters that are re-estimated are the state-space parameters. The evaluation spans the 96 months from January 2018 to December 2025.

The binding constraint is the publication lag. As mentioned the search data are

available essentially in real time, but TÜİK releases unemployment with a two-month delay(6-8 weeks to be more precise), so when the month- $t$  nowcast is formed the two most recent months,  $u_{t-1}$  and  $u_t$ , have not yet appeared. The information set is therefore

$$\mathcal{I}_t = \{u_s : s \leq t-2\} \cup \{\text{VPC}_{j,s} : s \leq t\}, \quad (9)$$

which we treat by letting  $u_{t-1}$  and  $u_t$  as missing in the measurement equation. For each month  $t$  in the evaluation window our procedure follows as.

1. Build  $\text{VPC}_{j,s}$  for  $s \leq t$  from the frozen pipeline.
2. We mask the lag, on the expanding sample up to month  $t$ , set  $u_{t-1}$  and  $u_t$  to missing, so unemployment is seen only through  $t-2$  while the factors are seen through  $t$ .
3. Estimate the state-space parameters by maximum likelihood,  $\hat{\theta}_t = \arg \max_{\theta} \ell(\theta; \mathcal{I}_t)$ , warm-starting at the previous month's estimate  $\hat{\theta}_{t-1}$ .
4. Then nowcast, run the Kalman filter and take the model-implied conditional mean of current unemployment,  $\hat{u}_t = \mathbb{E}_{\hat{\theta}_t}[u_t | \mathcal{I}_t]$ ; the contemporaneous factors  $\text{VPC}_{j,t}$  sharpen the filtered latent state  $\mu_t$  and thus the nowcast.
5. Once the official figure is released, record the realized error  $e_t = u_t - \hat{u}_t$ .

In a more visual sense our procedure can be seen in figure 8.

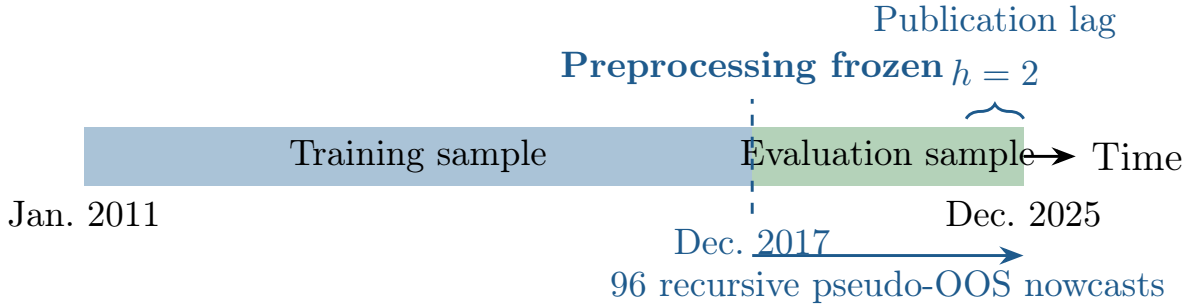


Figure 8: Recursive pseudo-out-of-sample evaluation design.

All benchmarks are handled identically: at each  $t$  an AR(1) and an AR(2) are re-estimated on  $\{u_s : s \leq t-2\}$  and we store their two-step-ahead forecasts of  $u_t$ .

We evaluate forecast accuracy via root mean squared error over the evaluation months  $\mathcal{E}$ .

$$\text{RMSE}(\mathcal{E}) = \sqrt{\frac{1}{|\mathcal{E}|} \sum_{t \in \mathcal{E}} e_t^2}, \quad (10)$$

We report RMSE over both the full window and over window excluding the COVID-19 period (April 2020–December 2021). To assess whether a Google-augmented model

improves on the benchmarkS we form the loss differential  $d_t = (e_t^{\text{AR}(1)})^2 - e_t^2$ , it will be positive when the candidate is the more accurate of the two, and test  $\mathbb{E}[d_t] = 0$  using the Diebold–Mariano statistic (Diebold and Mariano, 1995). Since the candidates nest the benchmark, we also report the Clark–West statistic (Clark and West, 2007), which corrects the comparison for the extra parameters estimated under the null. Our preferred statistic will be the CW statistic even though it may not be as robust as the Diebold Mariano test, Clark West corrects for the small sample bias DM has thus proving to be a better fit for our setup (small sample and nested model).

#### 4.4 Candidate Models

Every candidate we consider is a special case of the state-space models we introduced in the aforementioned sections; they are different only in which search factors they use and in how they treat the pandemic. We vary on the three lag orders: a first-order latent factor ( $p = 1$ ) of the transition equation, we fix this to 1 as AR(1) nearly outperforms other AR benchmarks thus it was logical and in some sense ad hoc to set the order of the transition equation to 1. Loadings on the current and two preceding values of the state ( $D = 2$ ) this comes naturally due to the publication lag. Finally a bridge with three possibilities of  $L = 1$ ,  $L = 2$ , and  $L = 3$ . We vary the COVID treatment with options of no dummy and fixed dummy. These choices keep the candidate count small, which matters at our sample size matters for avoiding overfitting.

A candidate is then selected by three ingredients. A candidate is then selected by three ingredients.

1. **Factor set  $\mathcal{S}$ .** A non-empty subset of the four rotated components: each of the four singletons  $\{\text{VPC}_k\}$  and each of the six pairs  $\{\text{VPC}_k, \text{VPC}_l\}$ .
2. **COVID treatment.** The pandemic term in the unemployment equation is either absent or a constant dummy  $\delta D_t^{\text{cov}}$
3. **Bridge depth  $L$ .** The number of lagged factors entering the unemployment equation directly,  $L \in \{0, 1, 2\}$ . Setting  $L = 0$  removes the bridge altogether, so the factors reach unemployment only through the shared latent state  $\mu_t$ ;  $L = 1$  and  $L = 2$  add one and two lagged factors, with  $L = 2$  spanning the full two-month gap between the last observed figure and the nowcast target.

We also carry a partial-least-squares variant for some of the models, in which a single supervised factor replaces the rotated components as a check on how the factors are extracted. Throughout, the AR(1), AR(2) and the state space without trends serve as our benchmarks.



## 4.5 Model Selection in Real Time

Reporting the single best-performing candidate over the full sample would tell us which specification won, but this would largely be within hindsight, not which one a forecaster could have chosen with restricted information. To make the comparison honest, we allow the data choose the model in real time, using only information available at each point in the evaluation.

The selection operates on the candidate models  $\mathcal{M}$  where each candidate  $m \in \mathcal{M}$  delivers a genuine recursive nowcast of  $\hat{u}_t^{(m)}$  from the design of. We re-select once a year on an expanding window. For each year  $y$  procedure follows as:

1. First rank, compute each candidate's out-of-sample RMSE over all evaluation months that precede year  $y$ ,

$$\mathcal{E}_{<y} = \{t : t < \text{January } y\}, \quad m_y^* = \arg \min_{m \in \mathcal{M}} \sqrt{\frac{1}{|\mathcal{E}_{<y}|} \sum_{t \in \mathcal{E}_{<y}} (e_t^{(m)})^2}. \quad (11)$$

2. Then we commit, select the best model  $m_y^*$  for the twelve months of year  $y$ , recording its nowcasts  $\hat{u}_t = \hat{u}_t^{(m_y^*)}$  for that year.

Our first evaluation year 2018 serves as the baseline on which the chosen model of 2019 is based, Hence this expanding window forecast is assessed over the seven years 2019 to 2025. Thanks to this any model that is chosen each year is purely from past out-of-sample performance, allowing us to not get contaminated by any look ahead bias.

## 5 Results

### 5.1 Out-of-Sample Nowcast Accuracy

We start our results with the accuracy of the models we searched in our grid. Table 2 collects the recursive pseudo-out-of-sample accuracy of the grid of twenty candidates, together with the non Google benchmarks, over the 96 from January 2018 to December 2025. We read the table with three questions in mind: which models actually are the best, where the additional forecasting accuracy comes from, and how reliable is the result.

The model that came out on top combines the two labor-relevant rotated factors and drops the pandemic dummy (VPC3+VPC4, no COVID). Its full-sample RMSE of 0.533 is 9.2% below the AR(1) benchmark, and it remains the best model again once the COVID months are removed (0.423). The improvement is significant under the Clark-West test ( $p = 0.018$ ), which is the right comparison here because the candidate nests the autoregressive(AR(1)) benchmark.

We observe the two patterns. Firstly almost all of the gain rides on a single factor. Every candidate that beats AR(1) contains VPC3, and the leading models simply pair it with a second factor or a COVID dummy; every specification built without VPC3, by contrast, sits at or above the benchmark. We find it reassuring that this is the very factor our cross-correlation analysis had flagged as leading unemployment by two months, the VPC that looked promising in sample is the one that actually delivers out of sample. The worst model in the grid is the VPC1+2 with a COVID dummy, this is a useful reminder of why we fixed the candidate set in advance: had we kept only the best-looking factor combinations, we would have overstated the gains and probably would have overfitted the values.

The second pattern is the one that raises certain questions which concerns the two tests of equal predictive ability telling different stories. The Clark-West statistic, which is powerful against a nested alternative, marks most of the VPC3-based models as significant improvements on AR(1). The Diebold–Mariano statistic, which assumes no nested models and is therefore more conservative, does not even select our headline model, it carries a two-sided DM  $p$ -value of 0.12. We take this to mean that the search factors genuinely carry predictive content, but that the content is modest relative to the sampling uncertainty in a ninety-six-month window. We confront this directly with the model confidence set in Section 5.4.

Finally, the Google-free benchmarks tell us where the gain actually comes from. Both no-Trends state-space models, which keeps all of the filtering/state-space machinery but removes out the search factor, are worse than AR(1), with RMSE ratios of 1.01 and 1.02. In other words, the state-space structure on its own buys us nothing: the entire improvement comes from the Google signal, we delve this deeper in Section 5.2.

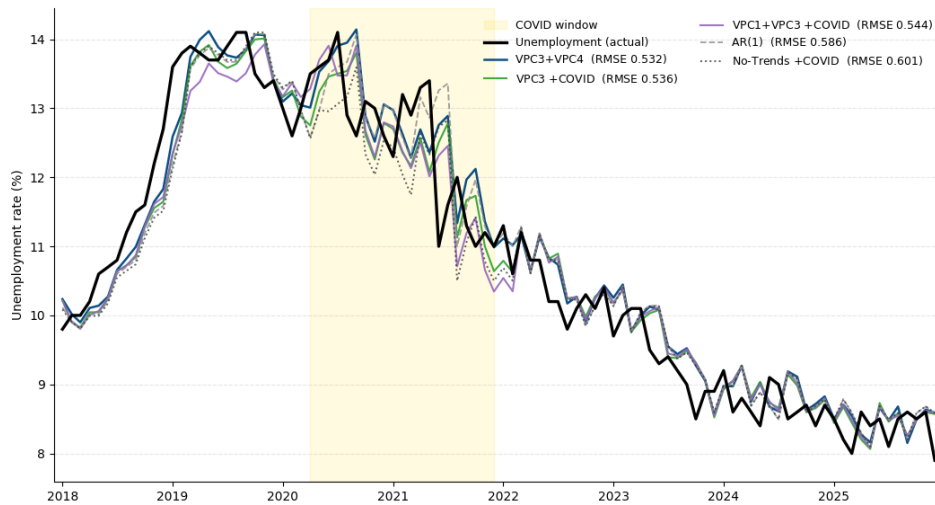


Figure 9: Fitted Values of the top 3 models vs benchmarks and the real unemployment series

Table 2: Out-of-sample nowcast accuracy, 2018–2025 ( $n = 96$ )

| Factors                                  | COVID | RMSE         | RMSE <sub>exC</sub> | RMSE/AR(1)   | $p_{DM}$     | $p_{CW}$       |
|------------------------------------------|-------|--------------|---------------------|--------------|--------------|----------------|
| <i>Panel A. Single factors</i>           |       |              |                     |              |              |                |
| VPC3                                     | dummy | 0.536        | 0.444               | 0.913        | 0.190        | 0.029**        |
| VPC3                                     | —     | 0.561        | 0.443               | 0.957        | 0.154        | 0.015**        |
| VPC4                                     | dummy | 0.564        | 0.476               | 0.961        | 0.618        | 0.071*         |
| VPC1                                     | dummy | 0.572        | 0.474               | 0.976        | 0.678        | 0.027**        |
| VPC4                                     | —     | 0.580        | 0.466               | 0.990        | 0.712        | 0.152          |
| VPC1                                     | —     | 0.588        | 0.469               | 1.003        | 0.934        | 0.143          |
| VPC2                                     | —     | 0.589        | 0.455               | 1.004        | 0.877        | 0.363          |
| VPC2                                     | dummy | 0.710        | 0.519               | 1.211        | 0.152        | 0.086*         |
| <i>Panel B. Factor pairs</i>             |       |              |                     |              |              |                |
| <b>VPC3+VPC4</b>                         | —     | <b>0.533</b> | <b>0.423</b>        | <b>0.908</b> | <b>0.118</b> | <b>0.018**</b> |
| VPC1+VPC3                                | dummy | 0.544        | 0.463               | 0.927        | 0.376        | 0.029**        |
| VPC3+VPC4                                | dummy | 0.546        | 0.446               | 0.931        | 0.502        | 0.040**        |
| VPC2+VPC3                                | —     | 0.551        | 0.433               | 0.940        | 0.095        | 0.009***       |
| VPC1+VPC3                                | —     | 0.558        | 0.456               | 0.952        | 0.164        | 0.013**        |
| VPC1+VPC4                                | dummy | 0.569        | 0.481               | 0.970        | 0.713        | 0.033**        |
| VPC1+VPC2                                | —     | 0.575        | 0.452               | 0.980        | 0.564        | 0.046**        |
| VPC2+VPC4                                | —     | 0.580        | 0.457               | 0.990        | 0.767        | 0.116          |
| VPC2+VPC3                                | dummy | 0.583        | 0.469               | 0.994        | 0.960        | 0.047**        |
| VPC1+VPC4                                | —     | 0.587        | 0.458               | 1.002        | 0.970        | 0.112          |
| VPC2+VPC4                                | dummy | 0.608        | 0.502               | 1.037        | 0.751        | 0.065*         |
| VPC1+VPC2                                | dummy | 0.807        | 0.541               | 1.377        | 0.090        | 0.092*         |
| <i>Panel C. Benchmarks (Google-free)</i> |       |              |                     |              |              |                |
| AR(1)                                    | —     | 0.586        | 0.469               | 1.000        | —            | —              |
| AR(2)                                    | —     | 0.606        | 0.447               | 1.034        | —            | —              |
| No-Trends SS                             | —     | 0.595        | 0.480               | 1.014        | —            | —              |
| No-Trends SS                             | dummy | 0.601        | 0.483               | 1.025        | —            | —              |

*Notes:* Recursive pseudo-out-of-sample nowcasts, January 2018–December 2025, two-month publication lag. RMSE is over the full window; RMSE<sub>exC</sub> excludes the COVID months (April 2020–December 2021). RMSE/AR(1) is the ratio to the AR(1) benchmark (values  $< 1$  beat it).  $p_{DM}$  is the two-sided Diebold–Mariano  $p$ -value (Harvey-corrected);  $p_{CW}$  is the one-sided Clark–West  $p$ -value, both against AR(1). All candidates fix the latent AR order, loading depth, and bridge depth at  $(p, D, L) = (1, 2, 2)$  and differ only in the factor set and COVID treatment. The headline model is in **bold**. No-Trends SS is the state-space filtering structure with the search factor removed.

Clark–West significance: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## 5.2 Decomposing the Gain: Filtering versus Google

A natural concern about our main result is that the improvement might be due to nature of the state-space model: the latent factor, the Kalman filter, the handling of the publication gap rather than to the google search data we introduced. To rule this out we add an intermediate model that keeps the state space machinery but removes the search signal. The no trends state space retains the structure (a latent AR(1) factor with measurement error, the same recursive design and filtered-state nowcast) but drops the Google terms entirely, thus  $u_t$  loads only on its own latent component. Comparing AR(1), the no-Trends model, and the top google model splits our gain into a filtering channel and a Google channel.

Table 3 reports all four models. The no-Trends models are worse than AR(1) with RMSE ratios of 1.01 and 1.02 so the filtering structure on its own contributes nothing, on the contrary it adds noise. The entire improvement appears only when the search factor is added. The top model lowers RMSE by 0.068, against a total gain over AR(1) of 0.054. In other words, the filtering channel is negative ( $-0.014$ ) and the Google channel ( $+0.068$ ) accounts for more than the whole of the headline's edge. A DM test isolating the search contribution against the no-Trends model with the same COVID DUMMY gives  $t = -1.81$  ( $p = 0.071$ ), which is significant at 90 percent.

Table 3: Decomposing the forecast gain: filtering versus Google

| Model                                                               | RMSE         | RMSE <sub>exC</sub> | RMSE/AR(1)   |
|---------------------------------------------------------------------|--------------|---------------------|--------------|
| AR(1)                                                               | 0.586        | 0.469               | 1.000        |
| No-Trends SS                                                        | 0.595        | 0.480               | 1.014        |
| No-Trends SS + COVID                                                | 0.601        | 0.483               | 1.025        |
| <b>VPC3+VPC4 (top model)</b>                                        | <b>0.533</b> | <b>0.423</b>        | <b>0.908</b> |
| <i>Channel decomposition</i> ( $\Delta$ RMSE vs AR(1), full window) |              |                     |              |
| Filtering structure (no Google)                                     |              | $-0.014$            |              |
| Google signal                                                       |              | $+0.068$            |              |
| Total gain                                                          |              | $+0.054$            |              |

*Notes:* Diebold- Mariano test of the top model against No-Trends SS + COVID (isolating the Google contribution):  $t = -1.81$ ,  $p = 0.071$ .

## 5.3 Real-Time Model Selection

The OOS comparison we did in Section 5.1 selects the best specification with hindsight. Why? Someone who is forecasting in real time does not have that information as she needs to choose a model from past performance alone. We therefore implement a look-ahead free rule of an expanding window. At the start of each year the twenty candidates

and the two AR benchmarks are ranked by their out-of-sample RMSE over all preceding months. We choose the leader for the next twelve months.

The selection results are quite stable, where the rule adopts VPC3+VPC4 (no COVID dummy) in every year from 2019 to 2025. The model a real-time forecaster would have committed to coincides with the best ex post specification, which something quite rare as ex post and ex ante is not expected to match each other. Here there is no hindsight gap, and the winner was already identifiable from 2018 and onwards.

Table 4: Real-time (expanding-selected) nowcast accuracy, 2019–2025

| Model                                                               | RMSE         | RMSE <sub>exC</sub> |
|---------------------------------------------------------------------|--------------|---------------------|
| <b>Expanding-selected</b>                                           | <b>0.546</b> | <b>0.423</b>        |
| AR(1)                                                               | 0.593        | 0.455               |
| AR(2)                                                               | 0.620        | 0.437               |
| Clark–West, Expanding-selected vs AR(1): $t = 1.77$ ( $p = 0.038$ ) |              |                     |

*Notes:* Test window 2019–2025; 2018 is the initial selection record.  
Clark–West is one-sided.

Table 4 reports the outcome. Over 2019–2025 the expanding-selected nowcast attains an RMSE of 0.546, against 0.593 for AR(1) and 0.620 for AR(2) when excluding the COVID months the numbers are 0.423 and 0.455. The improvement over AR(1) is significant under the Clark-West test with  $t = 1.77$ ,  $p = 0.038$  with the nested model being an AR(1). Figure 10 shows the real-time path the selected nowcasting model tracks the realized unemployment rate closely and departs from the AR(1) benchmark most visibly around the pandemic.

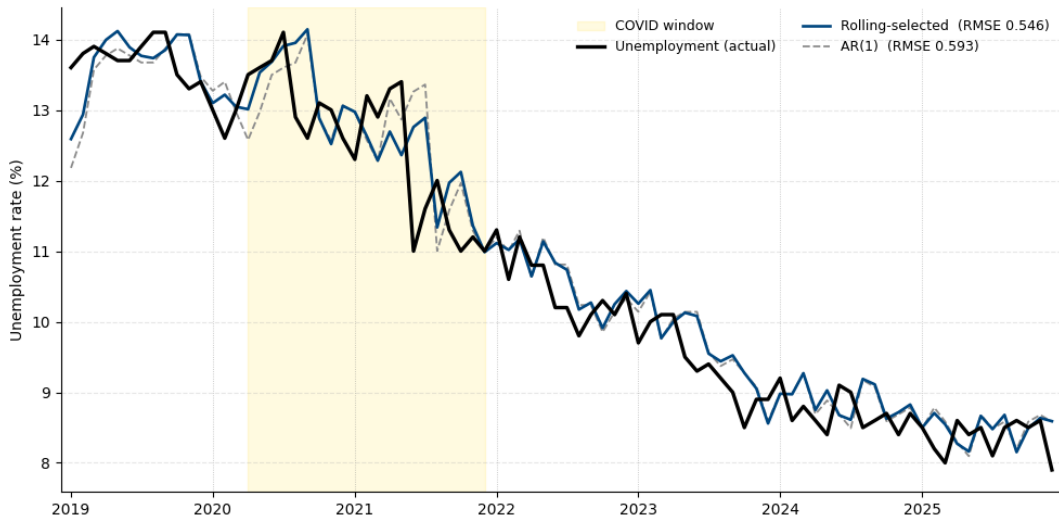


Figure 10: Real-time expanding-selected nowcast vs the real unemployment rate and AR(1)

## 5.4 The Model Confidence Set

So far we have compared models against AR(1) one pair at a time. With more than 20 specifications, we started to worry. Because we test a variety of models against the same benchmark, some of them are bound to look significant by chance alone. We wanted a way to check that takes the whole race between models seriously at once, so we turn to the model confidence set (MCS) of Hansen, Lunde, and Nason (2011), which returns the subset of models that cannot be told apart from the best at a chosen confidence level. We run it on the full pool—the twenty candidates together with the four Google-free benchmarks (AR(1), AR(2) and the two no-Trends models)—under squared-error loss, using the  $T_{\max}$  statistic and a moving-block bootstrap.

The answer is quite humbling and we would rather state it plainly than not discuss about them at all. At  $n = 96$  the procedure cannot distinguish anything, every single model, AR(1) and the no-Trends benchmarks included, sits inside the 90% confidence set on both the full and the ex-COVID windows (Table 5). Tightening the level to 75% does not help either—the smallest MCS  $p$ -value among all twenty-four models is still 0.26 on the full window and 0.40 ex-COVID. Our best model, VPC3+VPC4, does come out on top with an MCS  $p$ -value of 1.00, but the data simply will not reject the idea that plain AR(1) ( $p = 0.89$ ) nowcasts just as well.

Firstly this looked to us like it undercut everything we did before but it should not have a definite answer with two important caveats. Clark–West is a sharp, one-sided test of a single nested restriction, and it does pick up that the search factors carry predictive content. The MCS is a far more demanding joint test as it asks whether the models can be pulled apart from one another and with only eight years of monthly data it cannot. When we put these together, the picture becomes cleaner and more consistent. The Google signal is real, but it is not large enough, against the sampling noise in a sample this short, to select any one specification as the clear winner.

We have chosen to report this rather than keep it hidden because we think this is the honest way to proceed and the objective takeaway of the paper. GTAB-calibrated search data improve Turkish unemployment nowcasts by a margin that is modest, more concentrated in the first, (more turbulent) half of the sample, and statistically detectable, but not by one large enough to survive a comparison against a strong benchmark at the sample sizes we currently have.

Table 5: Model confidence set, 2018–2025

| Model                | RMSE  | MCS $p$ (full) | MCS $p$ (exC) |
|----------------------|-------|----------------|---------------|
| VPC3+VPC4 (best)     | 0.533 | 1.000          | 1.000         |
| VPC3 + COVID         | 0.536 | 0.951          | 0.935         |
| VPC1+VPC3 + COVID    | 0.544 | 0.951          | 0.802         |
| AR(1)                | 0.586 | 0.891          | 0.802         |
| No-Trends SS + COVID | 0.601 | 0.767          | 0.777         |
| AR(2)                | 0.606 | 0.774          | 0.935         |
| VPC2 + COVID         | 0.710 | 0.274          | 0.777         |
| VPC1+VPC2 + COVID    | 0.807 | 0.256          | 0.399         |
| 90% MCS              |       | all 24 models  | all 24 models |
| 75% MCS              |       | all 24 models  | all 24 models |

*Notes:* MCS of Hansen, Lunde, and Nason (2011) over the twenty candidates plus AR(1), AR(2), and the two no-Trends benchmarks (24 models). Squared-error loss,  $T_{\max}$  statistic, moving-block bootstrap (block length 3, 5,000 replications). A model is in the  $(1 - \alpha)$  set if its MCS  $p$ -value is at least  $\alpha$ . RMSE is full-window. The table lists representative models, you can find the full list in the [GitHub link](#).

## 5.5 When the Search Factors Help

The full-sample RMSE averages over very different macroeconomic conditions, and it contains very different time profiles. Table 6 splits the evaluation window into the pre-pandemic period (2018:01–2020:03), the COVID episode (2020:04–2021:12), and the post-pandemic period (2022:01–2025:12), and reports the RMSE of the best model alongside the benchmarks within each subperiod.

The advantage of the search-augmented model is far from uniform. It is largest before the pandemic- with an RMSE of 0.460 against 0.571 for AR(1), a reduction of around 19%-and remains substantial through the COVID episode. The period where the best model (0.810) and the COVID-augmented labor model (0.779) both clearly outperform AR(1) (0.886) and especially AR(2) (0.983). After 2022, however, the gain disappears entirely. The best model and AR(1) have essentially identical accuracy (0.401 versus 0.401), and no specification improves on the autoregressive benchmark. Figure 11 traces the same pattern continuously: the twelve-month rolling RMSE of the headline sits well below AR(1) through 2021 and then converges onto it.

Table 6: Out-of-sample RMSE by sub-period

| Model                         | Pre-COVID<br>( $n = 27$ ) | COVID<br>( $n = 21$ ) | Post-COVID<br>( $n = 48$ ) | Full<br>( $n = 96$ ) |
|-------------------------------|---------------------------|-----------------------|----------------------------|----------------------|
| <b>VPC3+VPC4 (Best Model)</b> | <b>0.460</b>              | 0.810                 | 0.401                      | <b>0.533</b>         |
| VPC3 + COVID                  | 0.521                     | <b>0.779</b>          | <b>0.394</b>               | 0.536                |
| AR(1)                         | 0.571                     | 0.886                 | 0.401                      | 0.586                |
| AR(2)                         | 0.527                     | 0.983                 | 0.396                      | 0.606                |
| No-Trends SS + COVID          | 0.597                     | 0.904                 | 0.404                      | 0.601                |

*Notes:* RMSE within each window. Pre-COVID: 2018:01–2020:03; COVID: 2020:04–2021:12; Post-COVID: 2022:01–2025:12. Best in each column in **bold**.

Our natural interpretation is that it is not that search data help specifically in turbulent times, but that their predictive content decays over the sample. The search signal is stronger in the first half and fades to nothing in the second. This is the forecasting counterpart of the declining contemporaneous correlation between the VPC’s and unemployment, which falls from around 0.74 in the pre-2018 estimation window to about 0.18 over the full sample. The modest full-window gain in Table 2 is therefore an average of two regimes: a genuine and sizeable edge through 2021, and nearly no gain thereafter. The likely sources of this instability can be the changing search behavior, the maturation of online job-search platforms, and the structural shocks of the period.

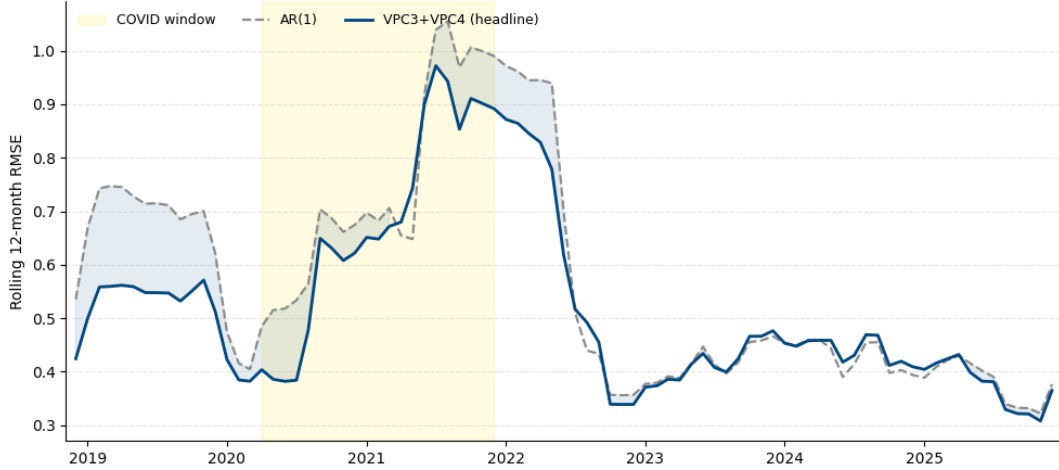


Figure 11: Twelve-month rolling RMSE of the headline model (VPC3+VPC4) and the AR(1) benchmark.

## 5.6 Robustness: Alternative Specifications

We have three modeling choices: a state-space shared factor (unemployment and the search factors load on a common latent state), a PCA-with-varimax dimension reduction



on the twenty-one keywords, and a year-on-year difference in preprocessing. We assess each of these selection choices by replacing one component at a time and re-running the identical recursive out-of-sample experiment. Table 7 collects the results, grouped by the dimension of model choice, every alternative uses the same keywords, so the comparisons isolate the modeling choice rather than the keyword selection.

At first we replace the shared-factor state space structure with three alternatives: PCs entering the latent transition equation, PCs entering only as exogenous regressors in the measurement equation, and a direct factor-augmented ARDL estimated by OLS without a Kalman filter. All three underperform the headline (0.581, 0.594, and 0.608 against 0.532), and the OLS ARDL is in fact worse than the AR(1) benchmark. The filtering structure that lets the model extract the latent state from the contemporaneous factors is doing real work. When we remove the filtering our gains evaporate, however most of these gains are due to the presence of google trends variables as demonstrated by prior sections.

Turning to dimension reduction, our next move is to replace the PCA with a supervised partial least-squares factor, (Kelly and Pruitt, 2015) in both the shared-factor and exogenous forms it yields 0.544 and 0.578, closer to the top model than the structural alternatives, but still it falls short of it. Not dealing with dimension reduction and entirely regressing unemployment on the raw keywords with elastic-net regularization (Zou and Hastie, 2005) is far worse (0.685). PLS fits the in-sample relationship more aggressively (due to the presence of the unemployment matrix) but does not generalize, consistent with the instability; PCA’s unsupervised factors are more robust out of sample. Finally, building the factors from deseasonalised levels rather than year-on-year differences produces the worst specification in the paper (0.759), confirming that the year-on-year transformation is what renders the search series informative.

Two readings reinforce our main results. No alternative improves on the best model on the full window. The structure we use of the shared-factor state space with PCA factors on year-on-year differenced data is still the best specification we examined. At the same time, several alternatives are competitive once the COVID months are excluded (the exogenous-PC, PLS-exog, and ARDL specifications all attain ex-COVID RMSE within 0.015 of the top model), the clear advantage of best model is again a full-window phenomenon rather than a uniform one.

Table 7: Robustness to alternative specifications, 2018–2025

| Specification                                                  | RMSE         | RMSE <sub>exC</sub> | RMSE/AR(1)   |
|----------------------------------------------------------------|--------------|---------------------|--------------|
| <b>Headline: shared-factor state space</b>                     | <b>0.532</b> | <b>0.423</b>        | <b>0.908</b> |
| <i>A. Alternative model structure (PCA, YoY factors fixed)</i> |              |                     |              |
| PCs in transition equation                                     | 0.581        | 0.445               | 0.991        |
| PCs as exogenous (measurement)                                 | 0.594        | 0.434               | 1.013        |
| Direct ARDL, OLS (no Kalman)                                   | 0.608        | 0.436               | 1.036        |
| <i>B. Alternative dimension reduction</i>                      |              |                     |              |
| PLS factor (state space)                                       | 0.544        | 0.460               | 0.927        |
| PLS-exog state space                                           | 0.578        | 0.437               | 0.985        |
| Raw keywords, elastic net (no PCA)                             | 0.685        | 0.518               | 1.168        |
| <i>C. Alternative preprocessing</i>                            |              |                     |              |
| Level factors (no YoY difference)                              | 0.759        | 0.463               | 1.295        |
| AR(1) benchmark                                                | 0.586        | 0.469               | 1.000        |
| AR(2) benchmark                                                | 0.606        | 0.447               | 1.034        |

*Notes:* Recursive out-of-sample RMSE, 2018–2025, two-month publication lag. All specifications use the same twenty-one keywords; each row changes one component relative to the headline. RMSE/AR(1) is the ratio to the AR(1) benchmark (values < 1 beat it). The reported figure for each alternative is its best-performing variant over the candidate lag/factor settings it considers.

## 5.7 Robustness: Bridge Depth

The best model fixes the bridge depth at  $L = 2$ , so that lagged factors enter the unemployment equation at times  $t - 1$  and  $t - 2$ . To check that this is not an arbitrary choice, we re-estimate every model with no bridge ( $L = 0$ , the factors reach unemployment only through the shared latent state), and a one-lag bridge ( $L = 1$ ), alongside the two-lag bridge ( $L = 2$ ) which we already did. Rest of the model remains fixed and the same as before. Table 8 reports our results.

The two-lag bridge is the most accurate of the three, so  $L = 2$  is the optimal model choice and not only a result implied by the publication lag. Interestingly enough a one-lag bridge is generally worse than no bridge at all. For the best models(indicated in bold) RMSE is 0.566 with no bridge, 0.617 with one lag, and 0.546 with two. The reason ties directly to the lead structure documented in Section 4.1: the labor factor leads unemployment by about two months, so a bridge that reaches only  $t - 1$  adds a noisy variable without adding the informative  $t - 2$  term. The forecasting performance we demonstrated with this exercise confirms the two-month lead and the two-month publication-lag structure.

Table 8: Robustness to bridge depth: out-of-sample RMSE, 2018–2025

| Factor set                      | $L = 0$ (no bridge) | $L = 1$ | $L = 2$      |
|---------------------------------|---------------------|---------|--------------|
| <i>Labor-relevant factors</i>   |                     |         |              |
| VPC3                            | 0.567               | 0.563   | <b>0.536</b> |
| VPC3+VPC4                       | 0.566               | 0.617   | <b>0.546</b> |
| VPC1+VPC3                       | 0.567               | 0.594   | <b>0.544</b> |
| VPC4                            | 0.570               | 0.604   | <b>0.564</b> |
| VPC1                            | 0.582               | 0.615   | <b>0.572</b> |
| <i>Weak (non-labor) factors</i> |                     |         |              |
| VPC2                            | <b>0.597</b>        | 0.710   | 0.710        |
| VPC2+VPC4                       | <b>0.572</b>        | 0.726   | 0.608        |

*Notes:* Recursive out-of-sample RMSE, 2018–2025, COVID-dummy specifications.  $L$  is the bridge depth (number of lagged factors entering the unemployment equation);  $L = 0$  retains only the shared-factor channel. AR(1) benchmark RMSE is 0.586. Best depth for each factor set in **bold**.

## 6 Discussion

Our results tell a consistent but deliberately modest story. GTAB-calibrated Google Trends carry real information about Turkish unemployment: the search-augmented models lower out-of-sample RMSE relative to the autoregressive benchmark, the Clark–West test confirms that the improvement is statistically significant, and the channel decomposition in Section 5.2 shows that the gain comes entirely from the search signal rather than from the state-space machinery that carries it. At the same time, the gain is modest, it is not constant over time, and—as the model confidence set makes clear—it is not large enough to single out any specification at the sample sizes we have.

The most striking pattern is the decay of the search signal over the evaluation window. The advantage is relatively large through 2021—both before the pandemic and during it—and essentially vanishes after 2022 (Section 5.5). We read this less as evidence that search data help specifically in turbulent times and more as a sign that the mapping from search behavior to unemployment is itself drifting. This is a familiar hazard for search-based predictors: Lazer et al. (2014) document how the much-celebrated Google Flu Trends model deteriorated as both search behavior and the platform itself evolved. Three forces plausibly contribute in our setting. First, search culture is changing: the data section already shows kariyer.net collapsing after 2013 as job seekers migrated to platforms such as LinkedIn, and queries that once tracked the labor market closely may simply have stopped doing so. Second, our preprocessing pipeline is frozen at December

2017, so the factor loadings cannot adapt to such shifts—a deliberate safeguard against look-ahead bias that comes at the cost of staleness. Third, the 2018–2021 window was unusually eventful, and a model fit to capture large swings may add little once the labor market settles into the calmer regime of 2022–2025, where the AR(1) benchmark is very hard to beat.

We have been intentionally objective about how strong the evidence is. The Clark–West test, which is the right tool for a single nested comparison, repeatedly flags the search factors as significant; the Diebold–Mariano test, which is more conservative, usually does not; and the model confidence set, which controls for the multiplicity of looking at two dozen specifications at once, cannot separate any model—search-augmented or not—from the field. None of these results contradicts the others. They are the expected readings of a real but small effect estimated on eight years of monthly data, and reporting all three rather than only the most favorable one is what we think makes the finding credible.

This picture sits comfortably within the recent literature. Like Henson (2025), who finds that the apparent value of Google Trends for US employment shrinks once forecasts are evaluated on genuinely real-time vintages, we find that an honest out-of-sample design dampens gains that look larger in sample—exactly the gap between our results and the strong in-sample improvements reported for Turkey by Chadwick and Şengül (2015). The state-dependence we document also echoes D’Amuri and Marcucci (2017), though in our data the dependence is on the calendar rather than on the business cycle alone. And the caveat that search captures job-seeking intent rather than unemployment itself (Baker and Fradkin, 2017), together with the sensitivity of search-based forecasts to who is online (Mulero and Garcia-Hiernaux, 2023), offers a natural explanation for why a fixed keyword panel loses traction as behavior evolves.

For a forecaster, the practical message is cautiously positive. A real-time rule that re-selects the model each year would have beaten the autoregressive benchmark over 2019–2025 by a small but statistically significant margin, at essentially no cost beyond collecting and calibrating the search data. Google Trends are therefore worth adding to a Turkish unemployment nowcasting toolkit—not as a decisive predictor, but as a cheap, real-time input whose value is largest exactly when official data are most stale, provided one remains alert to the drift in what search behavior measures.

## 7 Conclusion

We have asked whether GTAB-calibrated Google Trends can improve real-time nowcasts of the Turkish unemployment rate. Building a composite search index from 21 labor-market queries, embedding it in a state-space model with a latent labor-market factor, and evaluating the model in a strict recursive pseudo-out-of-sample exercise that masks

the two-month publication lag, we find that the search factors lower out-of-sample RMSE by up to about 9% over the full window and by nearly 20% before the pandemic. The improvement is significant under the Clark–West test, comes entirely from the Google signal rather than the filtering structure, and survives a fully real-time model-selection rule. It is, however, concentrated in the 2018–2021 portion of the sample, and not large enough to separate any single model from the field once multiplicity is taken into account.

Several limitations qualify these conclusions. The gain is small and time-limited, and the AR(1) benchmark is a tough competitor in calm periods. Search activity measures job-seeking intent rather than unemployment directly, so the mapping is noisy and, as our sub-period results suggest, unstable; our frozen preprocessing protects against look-ahead bias but cannot adapt to a shifting search culture. The evaluation rests on a single national series of only ninety-six monthly observations, which is precisely why the model confidence set is so uninformative, and the results are sensitive to how the COVID-19 episode is treated.

These limitations point to natural extensions. A longer or higher-frequency sample—for instance weekly Google Trends combined with the monthly target in a MIDAS–state-space hybrid (Ghysels, Sinko, and Valkanov, 2007)—would sharpen the tests that the present sample leaves inconclusive. Disaggregating to the regional or provincial level would multiply the effective sample and let search data speak to local labor markets. Allowing the preprocessing to update through time, rather than freezing it, would trade some look-ahead protection for the ability to track drift in search behavior. Finally, applying the same honesty, multiplicity-controlled design to other high-lag emerging-market labor markets would show whether the modest, decaying edge we document for Turkey is a general feature of search-based nowcasting or a strangeness of this episode.

## 8 Replication Code and Data

The Python files and data for both unemployment and calibrated trends(including the API code) can be found in the github link: <https://github.com/borderedhessian33/EC432-Google-Unemployment>.

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